A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 04

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1 Food:** cod, tuna, swordfish, sharks, turtles – extinct - dodo, great auk, passenger pigeon **1 Fashion:** leopard, snow leopard, tiger, ocelot fur – fur seals – crocodile / alligator leather – kingfisher, parrot, ostrich feathers **1 Entertainment / pets:** parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants **1 Furniture / ornaments:** mahogany, teak, ramin timber – ebony and ivory piano keys – turtle shell, shark teeth, coral and shells as souvenirs **1 Traditional medicines:** tiger (bones, claws), rhinoceros horn, seahorses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	5

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Threat to humans:** sharks, venomous snakes, crocodiles **1 Disease vectors:** malaria mosquitoes, tsetse flies **1 Predators of livestock:** wolves, puma, birds of prey, herons, seals **1** Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** High demand and prices make illegal trade / poaching profitable – laws are hard to enforce across borders **1** Belief in the medicine persists despite little scientific evidence – species become rarer and more valuable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** By-catch = 5.9 + 1.9 = 7.8 t – total = 8.4 t **1** 7.8 / 8.4 x 100 = 92.9% (accept 93%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.2	1 mark for the point below. **1** The nets have a very small mesh size so few non-target animals can escape **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mesh size** → net hole size, set so that young fish escape	1

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young **1** They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred **1** Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young – they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Impacts:** population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • **Catch quotas:** limit weight landed – work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • **Gear design:** mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag) – turtle exclusion devices – acoustic pingers for dolphins – curved hooks, decoys, sinkers and night setting to protect albatrosses – biodegradable / radio-tracked traps to reduce ghost fishing • **Restricted effort:** limits on boat size, engine power and days at sea – decommissioning – restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • **Evaluation:** enforcement at sea is difficult – some methods reduce catch or raise costs – quotas need reliable stock data – effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate – as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **quota** → a legal limit on the mass or number that may be caught	9
	• **mesh size** → net hole size, set so that young fish escape • **biodegradable** → broken down by decomposers **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Fishing's impacts go beyond depleting target species: k-selected species that breed late and produce few young collapse and recover slowly; by-catch of immature target fish, over-quota fish and non-commercial species is thrown back dead, with drift nets killing whales, dolphins and turtles, longlines drowning albatrosses and pair trawls porpoises; lost gear keeps ghost fishing; demersal trawls destroy sea fans, corals and seagrass; and removing any species alters its competitors, prey and predators.	
	Catch quotas limit the weight landed and work well for single-species pelagic shoals where fishing stops when the quota is reached, but in mixed demersal fisheries over-quota fish are discarded dead, so the quota does not protect them. Gear design tackles by-catch directly: mesh sizes and square-mesh escape panels let small fish out because, unlike diagonal mesh, they do not close under drag; turtle exclusion devices, acoustic pingers for dolphins, curved hooks, decoys, sinkers and night setting protect turtles, cetaceans and albatrosses; biodegradable and radio-tracked traps limit ghost fishing. Restricting effort - boat size, engine power, days at sea, decommissioning - reduces total pressure, and banning drift nets or demersal trawling on sensitive seabeds such as Rockall Bank protects habitats.	
	Effectiveness depends on enforcement at sea, which is difficult, on reliable stock data for quotas, on the species and fishery type, and on international cooperation because fish migrate; some measures cut catches or raise costs, and as local stocks fall fleets simply move further away. Gear modifications are the most effective per unit cost because they reduce damage without reducing legal catch; quotas and effort limits only work where they are enforced.	
	Weaker answer - Level 2 (5 marks)	
	Fishing damages the environment by overfishing, catching by-catch such as dolphins and damaging the seabed with trawling. Quotas limit how many fish can be caught so stocks can recover. Bigger mesh sizes let small fish escape so they can grow and breed. Turtle exclusion devices let turtles escape from nets. Some fishing methods can be banned in sensitive areas. No-take zones let fish breed. These methods help but they are hard to enforce at sea and some fishermen ignore them.	
	What makes the difference	
	• Level 2 covers impacts and methods soundly but without the mechanisms (why square mesh works, why quotas fail in mixed fisheries).	
	• Level 3 evaluates each method on its own terms and then compares them; Level 2's evaluation is a single general sentence.	
	• Named examples (Rockall Bank, albatross measures, ghost-fishing controls) give Level 3 the range the top band expects.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Images / maps of large areas – land-use change, deforestation, flooding **1** Vegetation density and surface temperature **1** Wind velocity, sea-surface altitude, wave height, oil pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	3
03.2	1 mark for the point below. **1** Microwaves pass through clouds (unlike visible light) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected **1** No need for researchers to visit inaccessible areas, reducing cost and disturbance – data are standardised across the whole area **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2
03.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night **1** Each tiger has a unique pattern of facial / body stripes **1** Images are compared with the database so individuals are recognised rather than counted twice **1** The number of different individuals recorded gives a minimum population – repeated captures give territory size, movements, lifespans and social groupings **1 Limitation:** only works for species with unique visible features **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (240 000 - 2 000) / 240 000 x 100 = 99.2% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Whales are k-selected:** slow growing, late maturing, one calf every few years, so populations recover very slowly **1** Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades **1** No regulation / open access to the ocean – whales were valuable (oil, spermaceti) and easily found **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Moratorium on commercial whaling since 1986 until populations recover **1** Total protection for certain species – whale sanctuaries **1** Limits on numbers and sizes taken – protection of suckling mothers and calves **1** Research into whale biology and threats (ship strikes, net entanglement) – exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since **1** Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses – whaling involves few nations and a single treaty body **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
05.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Impact depends on total population x per capita impact **1** With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added **1** Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Urban areas are supported by resources brought in from elsewhere – when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage **1** Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Climate:** warming increases evaporation, cloud and albedo – higher CO2 increases photosynthesis and sequestration – hydrological cycle: more evaporation gives more precipitation – populations: density-dependent factors return numbers to carrying capacity **up to 2** In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) **up to 2** Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **carrying capacity** → the largest population the area can support **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **biodiversity** → the variety of life: habitat, species and genetic diversity • **endemic** → found naturally only in that one place	1

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1 Rainforest:** high light, warm, wet all year, stable conditions for long periods, continuous productivity, many niches (stratification) **1 Antarctica:** very low temperatures, low precipitation, extreme seasonal light, 98% ice cover so almost no terrestrial life **1** Antarctic biodiversity is marine, supported by upwelling nutrients and the summer bloom – rainforest supports over half of terrestrial species on 10% of land **1** Both are important carbon stores / climate regulators but rainforest species are specialists vulnerable to small changes – Antarctic species adapted to extreme fluctuation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Less dead organic matter / leaf litter so less humus and slower soil formation – fewer nutrients returned **1** Loss of canopy and litter exposes soil to rain splash and wind **1** Decomposing roots no longer bind soil so erosion, gullying and landslides increase (especially on slopes) **1** Reduced organic matter and biota lower infiltration and water retention **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	4

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 4 points below. **1** Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind **1 Increased runoff:** more flooding after rain, lower flows in dry periods – sedimentation of rivers **1** Increased albedo of cleared land (reflects more sunlight) **1** Reduced carbon sequestration and loss of the carbon reservoir (CO_2 released by burning / decay) contributing to climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
07.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects **1** Loss of inter-species relationships (pollinators, seed dispersers) causing further declines **1** Loss of genetic resources / potential medicines before they are discovered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
09.1	1 mark for the point below. **1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Zebu tolerate heat but have a low milk yield – Ayrshire have a high milk yield but are not heat tolerant **1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Replacement of valuable animals that die with genetically identical individuals **1** Replacement of herds culled during a disease outbreak **1** Production of large numbers of individuals with desirable characteristics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them **1** In carefully controlled amounts these chemicals have beneficial physiological effects in humans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful **1** Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems **1 Hippopotamus:** hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment **1 Marine sponges:** proteins preventing graft rejection - organ transplants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Resources:** wood, fibres, oils, fuels, food – new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • **Biomimetics:** whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research – genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • **Ecosystem services:** atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination, seed dispersal, habitat provision • **Evaluation:** services go unnoticed because of dynamic equilibrium – unknown future value – economic vs moral / intrinsic arguments – conservation also needed for aesthetic and cultural reasons • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **biodiversity** → the variety of life: habitat, species and genetic diversity	9
	Model answer - Level 3 (9 marks)	
	Other species supply resources we use directly: wood, fibres such as cotton and wool, oils, fuels and food. Because most farmed species were domesticated long ago and are often kept where they are poorly adapted, wild species such as the drought-resistant morama bean, the yeheb tree that grows on poor soils, perennial maize that needs no ploughing, and bison that graze grassland too poor for cattle offer new, better-adapted food sources. Wild species also carry genetic resources: crop wild relatives in Vavilov centres hold genes for disease resistance and salt tolerance that inbred commercial varieties lack.	

Do not write outside the box

Question	Marking guidance	Marks
	Biomimetics copies adaptations into technology - humpback flipper tubercles in turbine blades, shark skin in friction-reducing coatings, termite-mound convection in naturally ventilated buildings, hollow bird bones in lightweight roofs. Medicines such as taxol, quinine, AZT and diosgenin came from wild species, and physiological research on marsupials, hippo skin and sponge proteins is helping with foetal development, burns and transplant rejection. Since only a tiny fraction of species has been studied, the unknown future value argues for conserving whole habitats.	
	Beyond resources, other species provide services we could not replace: regulation of atmospheric gases, evapotranspiration driving rainfall, the carbon, nitrogen and phosphorus cycles run by microbes, soil formation by decomposers, and pollination, seed dispersal and habitat provision. These are unnoticed precisely because they are in dynamic equilibrium, so their economic value is underestimated. On balance the resource and service arguments alone justify conservation, but they are strongest when combined with the intrinsic and aesthetic case, because many species with no known use still sustain the systems we depend on.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important because we get lots of things from other species. We get wood, cotton and food from plants and animals. Some medicines come from plants, like aspirin from willow trees. Animals can help design things, like Velcro was copied from seeds. Other species also do things for us like bees pollinating crops. If we lose species we might lose things that could be useful in the future. So we should conserve biodiversity.	
	What makes the difference	
	• Both answers cover resources, medicines, biomimetics and services, but Level 3 develops each with specific, accurate detail (which species, what property, why it matters).	
	• The question asks 'to what extent'. Level 3 weighs resource arguments against intrinsic value and reaches a reasoned conclusion; Level 1 asserts 'so we should conserve'.	
	• Level 3 explains *why* services are undervalued (dynamic equilibrium) - an explanation, not just a list.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Physical features:** mass / gravity retaining an atmosphere and giving pressure for liquid water – distance from the Sun and the Goldilocks zone – speed of rotation reducing temperature extremes – tilt producing seasons – magnetic field deflecting the solar wind – the Moon stabilising the orbit and producing tides • **Importance of liquid water:** solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • **Development of life:** Archaea at vents – photosynthesis – oxygen reacting with iron then accumulating – ozone layer allowing colonisation of land – aerobic organisms • **Life changing conditions:** carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases	25
	• **Methods of studying past conditions:** proxy data, isotope ratios, ice cores, Mg:Ca ratios – understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	

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Question	Marking guidance	Marks
11.2	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

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